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Wind power generator

Abstract

A wind power generator may include a wind turbine, a first motor generator configured to be driven by the wind turbine, a second motor generator, at least one mechanism configured to be driven by the second motor generator, a power split mechanism that couples a first shaft connected to the first motor generator, a second shaft connected to the second motor generator, and a third shaft to each other such that the first shaft, the second shaft, and the third shaft are configured to rotate differentially with respect to each other, and a rotation restricting mechanism configured to prohibit or restrict rotation of the third shaft.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority from Japanese Patent Application No. 2022-195117 filed on Dec. 6, 2022. The entire content of the priority application is incorporated herein by reference.

TECHNICAL FIELD

(2) The art disclosed herein relates to a wind power generator.

BACKGROUND ART

(3) Japanese Patent Application Publication No. 2013-53548 describes a wind power generator with a first motor generator driven by a wind turbine, a second motor generator, and at least one mechanism (a cooling pump) driven by the second motor generator. A first shaft connected to the first motor generator and a second shaft connected to the second motor generator are not connected to each other.

DESCRIPTION

- (4) In the configuration of Japanese Patent Application Publication No. 2013-53548, the cooling pump cannot be driven unless the power is supplied to the second motor generator for operation. Therefore, while power is generated by the first motor generator, the power is consumed by the second motor generator, which may cause waste in terms of energy efficiency. This is not limited to the case where a mechanism driven by the second motor generator is a cooling pump. The present disclosure provides a technology that can improve energy efficiency of a wind power generator.
- (5) A wind power generator disclosed herein may comprise a wind turbine, a first motor generator configured to be driven by the wind turbine, a second motor generator, at least one mechanism configured to be driven by the second motor generator, a power split mechanism that couples a first shaft connected to the first motor generator, a second shaft connected to the second motor generator, and a third shaft to each other such that the first shaft, the second shaft, and the third shaft are configured to rotate differentially with respect to each other, and a rotation restricting mechanism configured to prohibit or restrict rotation of the third shaft.
- (6) According to the above configuration, the power generated by the rotation of the wind turbine can be distributed to the first motor generator and the second motor generator. Therefore, the power generated by the rotation of the wind turbine can be used to rotate not only the first motor generator but also the second motor generator. This allows the at least one mechanism connected to the second motor generator to be driven by the power from the rotation of the wind turbine without supplying power to the second motor generator for operation. For example, when the rotation of the wind turbine generates surplus power under a strong wind condition, a part of the power can be distributed to the second motor generator. This can reduce the power consumed by the second motor generator when the at least one mechanism is driven. Thus, the energy efficiency of the wind power generator can be improved.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 schematically illustrates a wind power generator **50** of Embodiment 1.
- (2) FIG. 2 schematically illustrates a wind power generator **250** of Embodiment 2.
- (3) FIG. 3 schematically illustrates a wind power generator **350** of Embodiment 3.
- (4) FIG. 4 schematically illustrates a wind power generator **450** of Embodiment 4.
- (5) FIG. 5 schematically illustrates a wind power generator **550** of Embodiment 5.
- (6) In one or more embodiments, the at least one mechanism may comprise at least one of: a lubricating pump configured to supply lubricant to a part to be lubricated in the wind power generator; a cooling pump configured to supply coolant to a part to be cooled in the wind power generator; a cooling fan configured to supply cooling air to a part to be cooled in the wind power generator; a pitch angle changing mechanism configured to change a pitch angle of blades included in the wind turbine; and a yaw angle changing mechanism configured to change a yaw angle of a rotation axis of the wind turbine.
- (7) According to the above configuration, the power generated by the rotation of the wind turbine can be used for various purposes other than power generation.
- (8) In one or more embodiments, the wind power generator may further comprise a power supply circuit configured to supply power generated by the first motor generator to the second motor generator.
- (9) If the configuration in which an external power source supplies power to the second motor generator is employed, the power transmission loss may be relatively large. According to the above configuration, the power generated by the first motor generator can be supplied to the second motor generator. Therefore, the power transmission loss can be reduced.
- (10) In one or more embodiments, the rotation restricting mechanism may comprise at least one of:

a locking mechanism configured to selectively lock and unlock rotation of the third shaft; a clutch mechanism configured to selectively connect and disconnect the third shaft to and from the wind turbine; and a transmission mechanism that connects the third shaft to the wind turbine via a torque limiter.

(11) According to the above configuration, by at least one of the locking mechanism, the clutch mechanism, and the transmission mechanism, the way the power is split by the power split mechanism can be diversified.

(12) In one or more embodiments, the at least one mechanism may be connected to the third shaft and configured to be driven by the second motor generator via the power split mechanism. The at least one mechanism may be configured to work as the rotation restricting mechanism.

(13) According to the above configuration, the number of components in the wind power generator can be reduced because there is no need to provide a separate member such as the aforementioned locking mechanism.

Embodiment 1; Wind Power Generator **50**; FIG. **1**

(14) A wind power generator **50** in the present embodiment includes a hybrid unit **8**, a wind turbine **52**, a power conditioner **54**, a speed increasing gear **56**, and a locking mechanism **70**.

(15) The hybrid unit **8** is a power unit taken from a hybrid vehicle (not shown). The hybrid unit **8** includes a transaxle **6** and a power control unit **7**.

(16) The transaxle **6** includes a first motor generator **12** and a second motor generator **14**. The first motor generator **12** is a motor generator with a lower rated output than the second motor generator **14**. In the drawings, the first motor generator **12** may be illustrated as MG1 and the second motor generator **14** may be illustrated as MG2.

(17) The first motor generator **12** is connected to a first motor shaft **12a**. The first motor generator **12** can rotate the first motor shaft **12a** in response to power being supplied thereto. The first motor generator **12** can also generate power in response to the rotation of the first motor shaft **12a**.

(18) The second motor generator **14** is connected to a second motor shaft **14a**. The second motor generator **14** can rotate the second motor shaft **14a** in response to power being supplied thereto. The second motor generator **14** can also generate power in response to the rotation of the second motor shaft **14a**.

(19) The transaxle **6** further includes a planetary gear mechanism **16**. The planetary gear mechanism **16** connects an engine shaft **10a**, the first motor shaft **12a**, and the second motor shaft **14a** such that they are configured to rotate differentially with respect to each other. The planetary gear mechanism **16** can be called a power split mechanism.

(20) The planetary gear mechanism **16** includes a sun gear **16s**, a plurality of planetary gears **16p**, a planetary carrier **16c**, and a ring gear **16u**. The sun gear **16s** is connected to the first motor shaft **12a**. The plurality of planetary gears **16p** is arranged around the sun gear **16s** and is engaged with the sun gear **16s**. The planetary carrier **16c** rotatably supports the plurality of planetary gears **16p** and is connected to the engine shaft **10a**. The ring gear **16u** is located around the plurality of planetary gears **16p** and is engaged with the plurality of planetary gears **16p**. The ring gear **16u** is connected to the second motor shaft **14a** via a first speed reduction mechanism **18**. The ring gear **16u** is also connected to an axle **4a** via a second speed reduction mechanism **20** and a differential gear **21**. The axle **4a** is connected to a main shaft **52a** of the wind turbine **52** via a speed increasing gear **56**.

(21) The locking mechanism **70** is provided on the engine shaft **10a**. The locking mechanism **70** is configured to selectively lock and unlock the rotation of the engine shaft **10a**. When the rotation of the engine shaft **10a** is locked, the first motor shaft **12a** rotates with respect to the second motor shaft **14a** at a predetermined speed increase ratio. On the other hand, when the rotation of the engine shaft **10a** is unlocked, substantially no load is applied on the engine shaft **10a**. Therefore, the first motor shaft **12a** and the second motor shaft **14a** can rotate independently of each other. Hereafter, the state in which the rotation of the engine shaft **10a** is locked may be referred to as

“locked state”. The state in which the rotation of the engine shaft **10a** is unlocked may be referred to as “unlocked state”.

(22) The transaxle **6** further includes a mechanical oil pump **24**. The mechanical oil pump **24** is connected to the first motor shaft **12a** and is driven in response to the rotation of the first motor shaft **12a**. When the mechanical oil pump **24** is driven, the mechanical oil pump **24** circulates lubricating oil in the transaxle **6**. This allows each component of the transaxle **6** to be lubricated and cooled. A flow rate of the lubricating oil circulated by the mechanical oil pump **24** varies according to an amount of rotation of the first motor shaft **12a**.

(23) The power control unit **7** is integrated with the transaxle **6**. The power control unit **7** is electrically connected to an external power system **100** via the power conditioner **54**.

(24) The power control unit **7** includes a first inverter **26**, a second inverter **28**, a DC-DC converter **30**, and a control unit **31**. The control unit **31** is, for example, a power control unit (PCU). The first inverter **26** is electrically connected to the first motor generator **12**. The second inverter **28** is electrically connected to the second motor generator **14**. The first inverter **26**, the second inverter **28**, and the power conditioner **54** are electrically connected to each other.

(25) The first inverter **26** can convert AC power from the first motor generator **12** to DC power and supply it to the second inverter **28** and/or the power conditioner **54**. The first inverter **26** can also convert DC power from the second inverter **28** and/or the power conditioner **54** to AC power and supply it to the first motor generator **12**. The control unit **31** can control the operation of the first motor generator **12** via the first inverter **26**.

(26) The second inverter **28** can convert AC power from the second motor generator **14** to DC power and supply it to the first inverter **26** and/or the power conditioner **54**. The second inverter **28** can also convert DC power from the first inverter **26** and/or the power conditioner **54** to AC power and supply it to the second motor generator **14**. The control unit **31** can control the operation of the second motor generator **14** via the second inverter **28**.

(27) Temperature sensors **61** and **62** are provided on the first and second motor generators **12** and **14**, respectively. Temperature data outputted from the temperature sensors **61** and **62** is inputted to the control unit **31**. The control unit **31** can calculate the optimum value of a circulation flow rate of the lubricating oil based on the temperature data outputted from the temperature sensors **61** and **62**.

(28) The control unit **31** is electrically connected to the locking mechanism **70**. The control unit **31** can control the operation of the locking mechanism **70**. That is, the control unit **31** can switch the wind power generator **50** between the locked and unlocked states. Normally, the control unit **31** sets the wind power generator **50** in the locked state.

(29) (Operation of Wind Power Generator **50** in Locked State)

(30) When the wind turbine **52** rotates under wind power, the first motor shaft **12a** rotates and the second motor shaft **14a** rotates. In response to the rotation of the first motor shaft **12a**, power is generated by the first motor generator **12** and the mechanical oil pump **24** is driven. In addition, power is generated by the second motor generator **14** in response to the rotation of the second motor shaft **14a**.

(31) In the locked state, the mechanical oil pump **24** can be driven without supplying the power to the first motor generator **12** for operation. However, there can be a situation in which the circulation flow rate of the lubricating oil becomes excessive with respect to the optimum value or in which the circulation flow rate of the lubricating oil becomes insufficient with respect to the optimum value can be assumed. When such a situation is about to happen, the control unit **31** switches the wind power generator **50** to the unlocked state.

(32) (Operation of Wind Power Generator **50** in Unlocked State)

(33) When the wind turbine **52** rotates under wind power, the engine shaft **10a** rotates and the second motor shaft **14a** rotates. Power is generated by the second motor generator **14** in response to the rotation of the second motor shaft **14a**.

(34) The power control unit 7 supplies a part of the power generated by the second motor generator 14 to the first motor generator 12. This drives the first motor generator 12, and the first motor shaft 12a rotates. The power control unit 7 can adjust the amount of rotation of the first motor shaft 12a as it intends by adjusting the power supplied to the first motor generator 12. As a result, the power control unit 7 can drive the mechanical oil pump 24 so that the circulating flow rate of the lubricating oil becomes the optimum value.

Embodiment 2; Wind Power Generator 250; FIG. 2

(35) In the following, only the differences between a wind power generator 250 and the wind power generator 50 (see FIG. 1) are explained.

(36) In the present embodiment, a clutch mechanism 270 is provided instead of the locking mechanism 70 (see FIG. 1). The clutch mechanism 270 includes a clutch 272. An engine shaft 210a is provided instead of the engine shaft 10a (see FIG. 1). A speed increasing gear 256 is provided in place of the speed increasing gear 56 (see FIG. 1).

(37) The engine shaft 210a includes a first engine shaft 212a and a second engine shaft 214a. The first engine shaft 212a connects the clutch 272 to the speed increasing gear 256. The second engine shaft 214a connects the clutch 272 to the planetary gear mechanism 16.

(38) The speed increasing gear 256 connects the first engine shaft 212a and the main shaft 52a to each other so that the first engine shaft 212a rotates at a first speed increase ratio relative to the main shaft 52a. The speed increasing gear 256 also connects the axle 4a and the main shaft 52a to each other so that the axle 4a rotates at a second speed increase ratio relative to the main shaft 52a. The first speed increase ratio is greater than the second speed increase ratio.

(39) The clutch mechanism 270 is configured to selectively connect and disconnect between the first engine shaft 212a and the second engine shaft 214a. Hereafter, the state in which the first engine shaft 212a and the second engine shaft 214a are connected may be referred to as “clutch connected state”. The state in which the first engine shaft 212a and the second engine shaft 214a are disconnected may be referred to as “clutch disconnected state”. The control unit 31 can switch the wind power generator 250 between the clutch connected state and the clutch disconnected state. Normally, the control unit 31 sets the wind power generator 250 to the clutch connected state.

(40) (Operation of Wind Power Generator 250 in Clutch Connected State)

(41) When the wind turbine 52 rotates under wind power, the engine shaft 210a, the first motor shaft 12a, and the second motor shaft 14a rotate in conjunction with each other. In response to the rotation of the first motor shaft 12a, power is generated by the first motor generator 12 and the mechanical oil pump 24 is driven. In addition, power is generated by the second motor generator 14 in response to the rotation of the second motor shaft 14a.

(42) In the clutch connected state, the mechanical oil pump 24 can be driven without the operation of the first motor generator 12 to transmit the power thereto. However, there can be a situation in which the circulation flow rate of the lubricating oil becomes excessive with respect to the optimum value or in which the circulation flow rate of the lubricating oil becomes insufficient with respect to the optimum value can be assumed. When such a situation is expected to happen, the control unit 31 switches the wind power generator 250 to the clutch disconnected state.

(43) (Operation of Wind Power Generator 250 in Clutch Disconnected State)

(44) When the wind turbine 52 rotates under wind power, the second motor shaft 14a rotates. Power is generated by the second motor generator 14 in response to the rotation of the second motor shaft 14a.

(45) The power control unit 7 supplies a part of the power generated by the second motor generator 14 to the first motor generator 12. This drives the first motor generator 12, and the first motor shaft 12a rotates. The power control unit 7 can adjust the amount of rotation of the first motor shaft 12a as it intends by adjusting the power supplied to the first motor generator 12. As a result, the power control unit 7 can drive the mechanical oil pump 24 so that the circulating flow rate of the lubricating oil becomes the optimum value.

Embodiment 3; Wind Power Generator **350**; FIG. 3

(46) In the following, only the differences between a wind power generator **350** and the wind power generator **250** (see FIG. 2) are explained.

(47) In the present embodiment, a transmission mechanism **370** is provided instead of the clutch mechanism **270** (see FIG. 2). The transmission mechanism **370** includes a torque limiter **372**. The torque limiter **372** limits the torque generated between the first engine shaft **212a** and the second engine shaft **214a** at or below a predetermined value. For example, when the wind turbine **52** rotates excessively, the torque limiter **372** can reduce the torque inputted to the mechanical oil pump **24**. In this way, the torque limiter **372** can suppress the circulating flow rate of the lubricating oil from becoming excessive.

(48) (Operation of Wind Power Generator **350**)

(49) When the wind turbine **52** rotates under wind power, the engine shaft **210a**, the first motor shaft **12a**, and the second motor shaft **14a** rotate in conjunction with each other. In response to the rotation of the first motor shaft **12a**, power is generated by the first motor generator **12** and the mechanical oil pump **24** is driven. In the present embodiment, power generation by the first motor generator **12** is not performed.

(50) The power control unit **7** supplies a part of the power generated by the second motor generator **14** to the first motor generator **12**. This drives the first motor generator **12** and increases (or decreases) the rotation speed of the first motor shaft **12a**. The power control unit **7** can adjust the amount of rotation of the first motor shaft **12a** as it intends by adjusting the power supplied to the first motor generator **12**. In this way, the power control unit **7** can drive the mechanical oil pump **24** so that the circulating flow rate of the lubricating oil becomes the optimum value.

Embodiment 4; Wind Power Generator **450**; FIG. 4

(51) In the following, only the differences between a wind power generator **450** and the wind power generator **50** (see FIG. 1) are explained.

(52) In the present embodiment, a pitch angle changing mechanism **470** is provided instead of the locking mechanism **70** (see FIG. 1). The mechanical oil pump **24** is not connected to the first motor shaft **12a**. The mechanical oil pump **24** in the present embodiment is driven, for example, by an actuator (not shown).

(53) The pitch angle changing mechanism **470** is connected to the engine shaft **10a** and is driven in response to the rotation of the engine shaft **10a**. When the pitch angle changing mechanism **470** is driven, the pitch angle changing mechanism **470** changes a pitch angle of blades of the wind turbine **52**. Connecting the pitch angle changing mechanism **470** to the engine shaft **10a** applies a load on the engine shaft **10a**.

(54) (Operation of Wind Power Generator **450**)

(55) When the wind turbine **52** rotates under wind power, the engine shaft **10a**, the first motor shaft **12a**, and the second motor shaft **14a** rotate in conjunction with each other. The pitch angle changing mechanism **470** is driven in response to the rotation of the engine shaft **10a**. In addition, power is generated by the second motor generator **14** in response to the rotation of the second motor shaft **14a**. In the present embodiment, power generation by the first motor generator **12** is not performed.

(56) The power control unit **7** supplies a part of the power generated by the second motor generator **14** to the first motor generator **12**. This drives the first motor generator **12** and increases (or decreases) the rotation speed of the first motor shaft **12a**. The power control unit **7** can adjust the amount of rotation of the first motor shaft **12a** as it intends by adjusting the power supplied to the first motor generator **12**. By adjusting the amount of rotation of the first motor shaft **12a**, the power control unit **7** adjusts the amount of rotation of the engine shaft **10a**. This allows the power control unit **7** to operate the pitch angle changing mechanism **470** at the appropriate operating point.

Embodiment 5; Wind Power Generator **550**; FIG. 5

(57) In the following, only the differences between a wind power generator **550** and the wind

power generator **450** (see FIG. **4**) are explained.

(58) In the present embodiment, the pitch angle changing mechanism **470** is connected to the axle **4a** instead of being connected to the engine shaft **10a**. Further, the speed increasing gear **56** connects the engine shaft **10a** to the main shaft **52a** instead of connecting the axle **4a** to the main shaft **52a**.

(59) (Operation of Wind Power Generator **550**)

(60) When the wind turbine **52** rotates under wind power, the engine shaft **10a**, the first motor shaft **12a**, and the second motor shaft **14a** rotate in conjunction with each other. Power is generated by the first motor generator **12** in response to the rotation of the first motor shaft **12a**. In the present embodiment, power generation by the second motor generator **14** is not performed. The power control unit **7** supplies a part of the power generated by the first motor generator **12** to the second motor generator **14**. This drives the second motor generator **14** and increases (or decreases) the rotation of the second motor shaft **14a**. The power control unit **7** can adjust the amount of rotation of the second motor shaft **14a** as it intends by adjusting the power supplied to the second motor generator **14**. By adjusting the amount of rotation of the second motor shaft **14a**, the power control unit **7** adjusts the amount of rotation of the axle **4a**. This allows the power control unit **7** to operate the pitch angle changing mechanism **470** at the appropriate operating point.

Variants

(61) The position of the first motor generator **12** and the position of the second motor generator **14** may be interchanged with each other.

(62) Another power splitting mechanism (e.g., a differential gear) may be provided instead of the planetary gear mechanism **16**.

(63) Another mechanism may be provided in place of the mechanical oil pump **24**. For example, a cooling fan may be provided to supply cooling air to components of the transaxle **6**.

(64) Another mechanism may be provided instead of the pitch angle changing mechanism **470**. For example, a yaw angle changing mechanism configured to change a yaw angle of the main shaft **52a** of the wind turbine **52** may be provided.

(65) Two of the locking mechanism **70**, the clutch mechanism **270**, and the transmission mechanism **370** may be provided on the engine shaft **10a** (**210a**) at the same time. Alternatively, all of the locking mechanism **70**, the clutch mechanism **270**, and the transmission mechanism **370** may be installed on the engine shaft **10a** (**210a**) at the same time.

(66) The power control unit **7** may adjust the amount of rotation of the first motor shaft **12a** by adjusting the power generation torque applied to the first motor generator **12**.

(67) Power generation by the first motor generator **12** may not be performed. That is, the first motor generator **12** may function only as a prime mover to rotate the first motor shaft **12a**.

(68) Specific examples of the present invention have been described in detail, however, these are mere exemplary indications and thus do not limit the scope of the claims. The art described in the claims includes modifications and variations of the specific examples presented above. Technical features described in the description and the drawings may technically be useful alone or in various combinations, and are not limited to the combinations as originally claimed. Further, the art described in the description and the drawings may concurrently achieve a plurality of aims, and technical significance thereof resides in achieving any one of such aims.

Claims

1. A wind power generator, comprising: a wind turbine; a first motor generator mechanically connected to the wind turbine, wherein the first motor generator is driven by rotation of the wind turbine; a second motor generator; at least one mechanism configured to be driven by the second motor generator; a power split mechanism comprising: a ring gear, a plurality of planetary gears engaged with the ring gear, a planetary carrier rotatably supporting the plurality of planetary gears,

a sun gear engaged with the plurality of planetary gears, a first shaft connected to the first motor generator, wherein the ring gear is connected to the first shaft, a second shaft connected to the second motor generator, wherein the sun gear is connected to the second shaft, and a third shaft connected to the planetary carrier; and a rotation restricting mechanism configured to prohibit or restrict rotation of the third shaft, wherein the at least one mechanism comprises a pitch angle changing mechanism configured to change a pitch angle of blades included in the wind turbine, the pitch angle changing mechanism is connected to the third shaft and configured to be driven by the second motor generator via the power split mechanism, and the pitch angle changing mechanism is configured to work as the rotation restricting mechanism.

2. The wind power generator according to claim 1, further comprising a power supply circuit configured to supply power generated by the first motor generator to the second motor generator.
 3. The wind power generator according to claim 1, wherein the at least one mechanism comprises a cooling pump configured to supply coolant to a part to be cooled in the wind power generator.
 4. The wind power generator according to claim 1, wherein the at least one mechanism comprises a cooling fan configured to supply cooling air to a part to be cooled in the wind power generator.
 5. The wind power generator according to claim 1, wherein the at least one mechanism comprises a yaw angle changing mechanism configured to change a yaw angle of a rotation axis of the wind turbine.
 6. The wind power generator according to claim 1, wherein the rotation restricting mechanism comprises a locking mechanism configured to selectively lock and unlock rotation of the third shaft.
 7. The wind power generator according to claim 1, wherein the rotation restricting mechanism comprises a clutch mechanism configured to selectively connect and disconnect the third shaft to and from the wind turbine.
 8. The wind power generator according to claim 1, wherein the rotation restricting mechanism comprises a transmission mechanism that connects the third shaft to the wind turbine via a torque limiter.
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